

LIST OF 50 PYTHON INTERVIEW QUESTIONS,

broken down by category from core basics to Object-Oriented Programming (OOP) and programming logic.

Part 1: Python Basics & Core Concepts

Syntax & Data Types

1. What are the built-in, mutable, and immutable data types in Python?
2. What is the difference between a list and a tuple? When would you choose one over the other?
3. How does memory management work in Python? What is the role of the Garbage Collector?
4. What is the difference between `is` and `==` operators?
5. What are Python decorators, and how do you create one?
6. Explain the concept of PEP 8. Why is it important?
7. What is a lambda function? Give an example of where it is useful.
8. How do `pass`, `continue`, and `break` statements differ in loops?
9. What is the purpose of the `__init__` file in a Python package folder?
10. Explain the difference between shallow copy and deep copy.

Data Structures & Built-in Functions

11. What is a list comprehension? Convert a standard `for` loop that squares numbers into a list comprehension.
12. How do dictionaries store data efficiently in Python? (Explain hashing basics).
13. What is the difference between `dict.get('key')` and `dict['key']`?
14. What are `*args` and `kwargs` in function definitions? How are they used?
15. What is a generator? How does it differ from a normal function, and what is the use of the `yield` keyword?

16. **How does the `zip()` function work? Provide a quick use case.**
17. What is the difference between `range()` and `xrange()`? *(Note: relevance in Python 2 vs 3).*
18. **How can you remove duplicates from a list while maintaining or not maintaining its order?**
19. **What are set operations in Python? How do you find the intersection and union of two sets?**
20. **Explain how exception handling works using `try`, `except`, `else`, and `finally`.**

Part 2: Object-Oriented Programming (OOP) Concepts

Core Pillars

21. **What are the four main pillars of OOP? Briefly explain each in the context of Python.**
22. **How do you achieve Encapsulation in Python, given that it doesn't have strict private or protected keywords?**
23. **What is the purpose of the `super()` function? Give an example in inheritance.**
24. **Does Python support multiple inheritance? If so, how does it resolve method name conflicts? (Explain MRO / Method Resolution Order).**
25. **What is Polymorphism? How can you demonstrate it using function overriding and function overloading in Python?**
26. **What is the difference between Abstract Classes and Interfaces in Python? How do you create an abstract class?**

Methods, Attributes & Dunder Methods

27. **What is the difference between an Instance method, a Class method (`@classmethod`), and a Static method (`@staticmethod`)?**
28. **What are "dunder" (double underscore) or magic methods? Name three common ones and their purposes.**
29. **What is the difference between `__str__` and `__repr__`?**

- 30. What are slots (`__slots__`) in Python classes, and when should you use them?
- 31. Explain the concept of Name Mangling in Python. How is it triggered?
- 32. How can you make a Python object callable like a function?
- 33. What is the difference between a class attribute and an instance attribute?
- 34. How does Python's property decorator (`@property`) work? Why is it useful?
- 35. What is a metaclass in Python? When would you ever need to use one?

Part 3: Basic Programming & Logic Questions

(These are designed to test coding logic, string manipulation, and basic algorithmic thinking).

- 36. Write a Python program to check if a given string is a palindrome.
- 37. Write a program to print the Fibonacci series up to n terms using both recursion and iteration.
- 38. How do you swap two variables in Python without using a third temporary variable?
- 39. Write a script to count the frequency of each character in a string using a dictionary.
- 40. How do you reverse a list or a string using slicing? Explain the syntax `[::-1]`.
- 41. Write a function to find the factorial of a number using recursion.
- 42. Given a list of numbers, write a program to find the second largest number.
- 43. Write a Python program to check if a given year is a leap year.
- 44. How do you check if a number is prime in Python?
- 45. Write a program to merge two sorted lists into one sorted list.
- 46. How do you find the sum of all elements in a nested list (e.g., `[1, [2, 3], [4, [5]]]`)?
- 47. Write a Python snippet to find the intersection of two lists.
- 48. How do you check if two strings are anagrams of each other?
- 49. Write a program to find the first non-repeating character in a string.
- 50. Given a list of integers, write a program to move all zeros to the end of the list while maintaining the relative order of the non-zero elements.